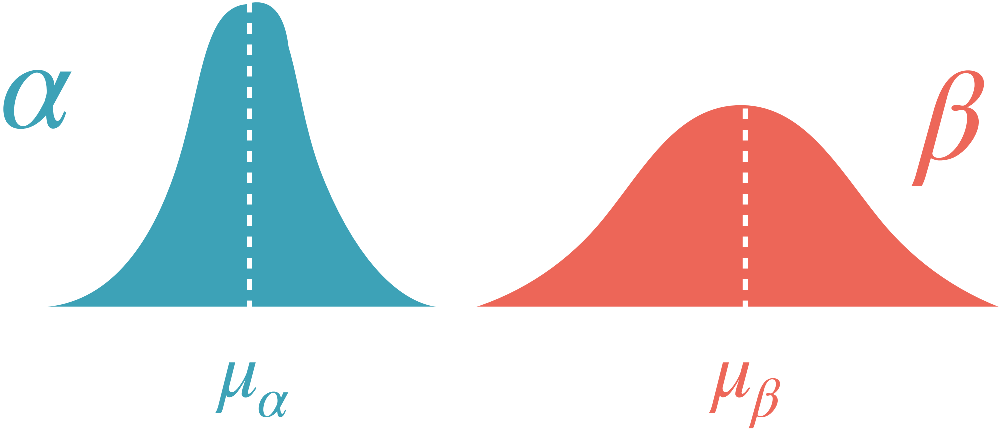
$$\alpha = \mathcal{N}(\mu_\alpha, \sigma_\alpha^2)$$

$$\beta = \mathcal{N}(\mu_\beta, \sigma_\beta^2)$$

$$\frac{\mathrm{d}\alpha}{\mathrm{d}x} = \frac{1}{\sigma_\alpha\sqrt{2\pi}}\exp\left(-\frac{(x-\mu_\alpha)^2}{2\sigma_\alpha^2}\right)$$

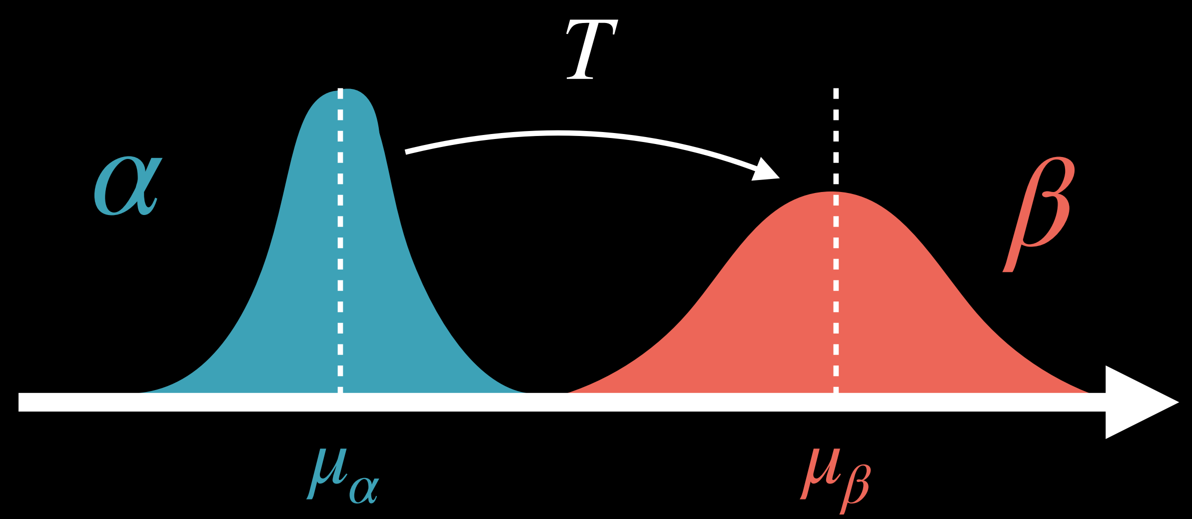
Property: An affine transformation of a Gaussian r.v. is also Gaussian



α
β
μ_α
μ_β

OT between 1D Gaussians

$\alpha = \mathcal{N}(\mu_\alpha, \sigma_\alpha^2) \quad \beta = \mathcal{N}(\mu_\beta, \sigma_\beta^2)$



Exercise: Show that for quadratic cost opt T (1) exists and (2) derive its form

Solutions:

1. Both measures have densities, e.g., $\dfrac{\mathrm{d}\alpha}{\mathrm{d}x} = \dfrac{1}{\sigma_\alpha\sqrt{2\pi}}\exp\left(-\dfrac{(x-\mu_\alpha)^2}{2\sigma_\alpha^2}\right)$. By Brenier, T exists and is unique

2. Note that $T(x) = \dfrac{\sigma_\beta}{\sigma_\alpha}(x-\mu_\alpha) + \mu_\beta$ does what we want [$T(X)$ has the right mean/variance]. Also, note that T is the derivative of $\varphi(x) = \dfrac{\sigma_\beta}{2\sigma_\alpha}(x-\mu_\alpha)^2 + \mu_\beta x$ and that φ is convex. By Brenier's theorem, T is *the* optimal transport map.

OT between N-D Gaussians

$\alpha = \mathcal{N}(\mu_\alpha, \Sigma_\alpha)$ $\qquad$ $\beta = \mathcal{N}(\mu_\beta, \Sigma_\beta)$